AD 502 – Machine Learning

Pre-Reading Material – Lecture 34

Basic Ensemble Techniques: Averaging

Introduction

Averaging is a basic ensemble learning technique used primarily for regression problems. It combines numerical predictions from multiple machine learning models to generate a single final prediction.

Instead of relying on one model's estimate, averaging considers predictions from several models. This can improve stability and reduce prediction errors when the models have complementary strengths.

Working Principle

The averaging technique follows these steps:

1. Train multiple regression models using the training dataset.
2. Provide the same test observation to each trained model.
3. Collect the numerical predictions generated by the models.
4. Calculate the average of the predictions.
5. Use the resulting average as the final prediction.

Example of Averaging

Suppose three models are trained to predict electricity demand.

Model	Predicted electricity demand
Linear Regression	100 MW
Decision Tree	110 MW
Random Forest	105 MW

The final prediction is obtained by averaging the three values:

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$$Final\ Prediction = \frac{100 + 110 + 105}{3}$$

The final predicted electricity demand is 105 MW.

Types of Averaging

A. Simple Averaging

In simple averaging, all models receive equal importance. Their numerical predictions are combined using the arithmetic mean.

This method is straightforward and works well when the models have reasonably comparable predictive performance.

B. Weighted Averaging

In weighted averaging, different models are assigned different weights according to their reliability, performance, or importance.

For example, suppose two models predict electricity demand as 100 MW and 110 MW. If the first model receives a weight of 0.7 and the second receives a weight of 0.3, the final prediction is:

$$(100 \times 0.7) + (110 \times 0.3) = 103MW$$

The first model has greater influence because it receives a higher weight.

For a weighted average, the weights are generally non-negative and sum to one.

Advantages of Averaging

- Simple to implement and understand.
- Can reduce the variability of predictions.
- Combines the strengths of multiple regression models.
- Can improve generalization when the models make different errors.
- Weighted averaging allows more reliable models to have greater influence.

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Limitations of Averaging

- Models with poor predictions can reduce overall performance.
- Simple averaging gives equal importance to all models.
- Similar models may provide little additional benefit.
- Weighted averaging requires an appropriate method for selecting weights.
- Averaging does not guarantee improved accuracy.

Applications

Averaging is commonly used in:

- Electricity demand forecasting.
- Weather prediction.
- Sales and price forecasting.
- Renewable energy generation forecasting.
- Temperature prediction.
- Other numerical prediction tasks.

For example, in renewable energy forecasting, predictions from Linear Regression, Decision Tree, and Random Forest models can be averaged to estimate future solar power generation.

Summary

Averaging is an ensemble technique that combines numerical predictions from multiple regression models. Simple averaging gives equal importance to all models, whereas weighted averaging assigns different levels of importance. The technique is particularly useful when models have complementary prediction errors and can provide a more stable final estimate.

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